

Lecture 8: Numerical optimization and integration

ECON 7510

Cornell University

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Numerical optimization

Motivating example: Logit model of demand

Suppose an employer offers its employees two health plans, $a = 0, 1$. The cost of both is the same, but they differ in the specifics of their coverage.

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- So, conditional on x , the share of employees who choose $a = 1$ is

$$\begin{aligned} p(x) &\equiv \Pr(a_i = 1 \mid x_i = x) \\ &= \Lambda(x' \beta) \\ &= \frac{\exp(x' \beta)}{1 + \exp(x' \beta)} \end{aligned}$$

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Estimation problem

- Suppose we observe (x_i, a_i) for $i = 1, \dots, l$.
- Estimate by maximum likelihood:

$$\begin{aligned}\hat{\beta}_{\text{MLE}} &= \operatorname{argmax}_{\beta} \prod_i p(x_i; \beta)^{a_i} (1 - p(x_i; \beta))^{1-a_i} \\ &= \operatorname{argmax}_{\beta} \sum_i \underbrace{a_i \log(x_i; \beta) + (1 - a_i) \log(1 - p(x_i; \beta))}_{\ell(x_i; \beta)}\end{aligned}$$

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$$\begin{aligned}\nabla_{\beta} \ell(x_i; \beta) &= \left[\frac{a_i}{p(x_i; \beta)} - \frac{1 - a_i}{1 - p(x_i; \beta)} \right] \nabla_{\beta} p(x_i; \beta) \\ &= \frac{a_i - p(x_i; \beta)}{p(x_i; \beta) (1 - p(x_i; \beta))} \nabla_{\beta} p(x_i; \beta)\end{aligned}$$

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- **Question:** Can we analytically solve FOC for $\hat{\beta}_{\text{MLE}}$?

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- **Question:** What is a geometric interpretation of the gradient that might be useful when thinking about optimization?

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- **Question:** Can we analytically solve FOC for $\hat{\beta}_{\text{MLE}}$?
- **Question:** What is a geometric interpretation of the gradient that might be useful when thinking about optimization? The gradient points in the direction of (locally) steepest ascent.
- Key idea: if start at some β and want to move toward $\hat{\beta}_{\text{MLE}}$, the gradient is a good direction to go. 3/16

Gradient Descent: A simple algorithm for numerical optimization

$$\min_{\theta} f(\theta)$$

Algorithm:

1. Start at some θ_1 .
2. Let $\theta_{k+1} = \theta_k - \eta \nabla f(\theta_k)$.
3. Repeat step 2 until $\nabla f(\theta_k) \approx 0$.

η is the *learning rate*. Gradient may give a good direction locally, but a bad direction more globally.

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Applying gradient descent

See `L8.py`

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 - Introducing *stochasticity*, e.g., SGD
 - What about **constrained** optimization problems?

Numerical constrained optimization

$$\min_{\theta} f(\theta) \text{ subject to } g(\theta) \leq 0$$

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- As it turns out, what works much better in practice is **Augmented Lagrangian** method.

Numerical constrained optimization

Augmented Lagrangian method

$$\mathcal{L}(\theta, \lambda, \kappa) = f(\theta) + \lambda'g(\theta) + \frac{\kappa}{2}g(\theta)'g(\theta)$$

- Choose initial parameters θ_0
- While not converged:
 - Primal update: $\theta \leftarrow \arg \min_{\theta} \mathcal{L}(\theta, \lambda, \kappa)$
 - Dual update: $\lambda \leftarrow \lambda + \kappa g(\theta)$

κ is a hyperparameter that has to be chosen in advance.

Numerical constrained optimization

A lazier (and better) method!

- `torch` and `scipy.optimize` work well together, and the latter can handle constrained problems.
- See `scipy-constrained-optimization-pytorch-example.py` for an example implementation.

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$$\bar{\epsilon} = \mathbb{E} \left[\frac{\partial p(x)}{\partial x} \frac{x}{p(x)} \right] = \int \frac{\partial p(x)}{\partial x} \frac{x}{p(x)} \phi(x) dx$$

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$$\begin{aligned}\frac{\partial p(x)}{\partial x} &= \frac{\partial}{\partial x} \frac{\exp(x'\beta)}{1 + \exp(x'\beta)} \\ &= p(x) (1 - p(x)) \beta\end{aligned}$$

$$\begin{aligned}\epsilon(x) &= \frac{\partial p(x)}{\partial x} \frac{x}{p(x)} \\ &= (1 - p(x)) \beta x\end{aligned}$$

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2. Gauss-Hermite quadrature: Exploit a special property of the Normal distribution.

Gauss-Hermite quadrature

Key idea: By evaluating the function at a small number of (carefully chosen) points, we can get enough info about the function's behavior to approximate the integral very well.

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$$\begin{aligned}\mathbb{E}[h(x)] &= \int_{-\infty}^{\infty} h(x) \phi\left(\frac{x-\mu}{\sigma}\right) dx \\ &\approx \frac{1}{\sqrt{\pi}} \sum_j w_j h(\sqrt{2}\sigma x_j + \mu)\end{aligned}$$

If we choose $w_1, \dots, w_n$ and $x_1, \dots, x_n$ wisely, we can show that this approximation is very good

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If we choose $w_1, \dots, w_n$ and $x_1, \dots, x_n$ wisely, we can show that this approximation is very good; in fact, it is *exact* if h is a polynomial of degree $2n - 1$ or less.

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More general case

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Preliminary question: Suppose you can only take draws from a Normal distribution. Is there a way you can obtain random draws from other continuous distributions?

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Suppose $x \sim F$ and $z \sim \Phi$. Note that $F(x), \Phi(z)$ are both distributed $U[0, 1]$. It follows that

$$F^{-1}(\Phi(z)) \sim F$$

has the same distribution as x .

Gauss-Hermite quadrature

More general case

Change of variable:

$$\begin{aligned}\mathbb{E}[h(x)] &= \int h(x) f(x) dx \\ &= \int h(F^{-1}(\Phi(z))) f(F^{-1}(\Phi(z))) \frac{dx}{dz} dz \\ &= \int h(F^{-1}(\Phi(z))) \phi(z) dz \\ &= \mathbb{E}[h(F^{-1}(\Phi(z)))] \\ &\approx \frac{1}{\sqrt{\pi}} \sum_j w_j h(F^{-1}(\Phi(\sqrt{2}z_j)))\end{aligned}$$

(The third step uses the Inverse Function Theorem.)

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Question: When doing this kind of change of variable, how should that affect our choice of the number of quadrature points?